

Adaptive Sparse Attention for Long-Document Understanding

A. Kumar¹, B. Okafor², C. Lindqvist^{1,3}
¹Dept. of CS, Univ. A; ²Lab B; ³Institute C

Abstract. We propose a sparse attention mechanism whose complexity is $O(n \log n)$ and show it improves accuracy on five benchmarks. **Abstract.** We propose a sparse attention mechanism whose complexity is $O(n \log n)$ and show it improves accuracy on five benchmarks. **Abstract.** We propose a sparse attention mechanism whose complexity is $O(n \log n)$ and show it improves accuracy on five benchmarks.

1. Introduction

Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input.

2. Method

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Attention is computed as: $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$ (1)

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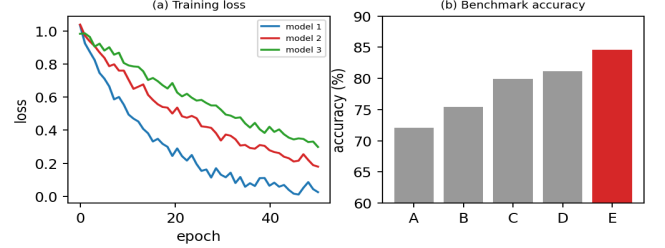


Fig. 1: (a) training loss curves, (b) benchmark accuracy (red = ours).

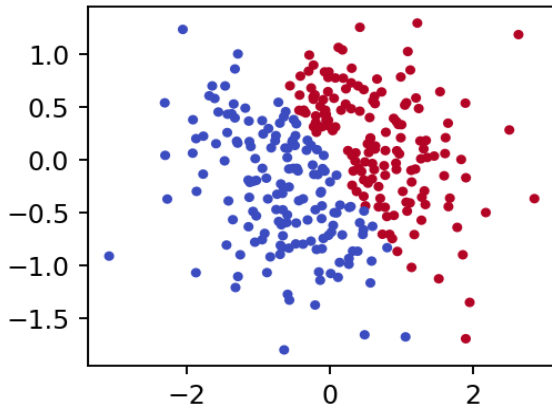
3. Experiments

Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input. Long documents require models that scale gracefully with sequence length. Prior work [1,2] relies on fixed sparsity patterns, which fail when salient tokens are distributed irregularly across the input.

Table 1: Results (booktabs style, no vertical lines).

Model	Params	Acc.	F1
Baseline	110M	72.1	70.4
Longformer	149M	75.4	73.9
BigBird	128M	79.9	78.2
Ours	112M	84.6	83.1

Fig. 2: Latent space



4. Conclusion

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